



Real-time hass avocado ripeness detection using close-range UAV imagery and YOLO-based deep learning

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ABSTRACT

Rapid and accurate assessment of Hass avocado ripeness is essential for improving harvest planning, optimizing picking time, increasing productivity, and reducing postharvest losses. This study presents a real-time system for Hass avocado ripeness detection using close-range UAV imagery and YOLO-based deep learning models. A UAV equipped with a camera captured field images, which were transmitted in real time to a processing station through SCRCPY for fruit detection and ripeness classification. The proposed approach consists of two sequential stages. First, a YOLOv8 object detection model was trained using a dataset of 340 manually annotated images generated through real-world adaptation and labeled in Roboflow. The detector achieved a precision of 0.7951, a recall of 0.7849, an mAP@0.5 of 0.8388, and an mAP@0.5:0.95 of 0.6308, demonstrating reliable fruit localization under field conditions. Second, a YOLOv8 classification model was trained on a dataset of 20,231 cropped fruit images derived from an initial collection of 14,710 images categorized into five ripeness stages (unripe, midripe, ripe, fullripe, and overripe). The classification model achieved an accuracy of 0.79, a macro-average F1-score of 0.77, a weighted-average F1-score of 0.78, a Top-1 accuracy of 0.78, and a Top-5 accuracy of 1.00, confirming its effectiveness for ripeness assessment. The complete system was evaluated under real agricultural conditions with variable illumination and partial fruit occlusion, demonstrating that the integration of close-range UAV imagery and deep learning provides an effective tool for automated avocado ripeness assessment in precision agriculture. The proposed approach was evaluated using close-range UAV imagery acquired in a commercial orchard; therefore, additional validation under different orchards, environmental conditions, and acquisition scenarios is recommended to further assess its generalization capability.

1. Introduction

Detecting Hass avocados is a challenge in agriculture because distinguishing their ripening stages is difficult, and the presence of leaves further complicates the task. Deep learning has become an essential technology for solving complex problems in various fields, and its applications have been widely adopted in agriculture through image-based models for fruit detection, ripeness classification, and crop monitoring. These technologies contribute to improving field operations by accelerating identification processes, reducing operating time, and supporting decision-making. Monitoring the maturity level of Hass avocados using traditional methods presents operational limitations and high inspection costs due to the need for qualified personnel, manual data collection, and time-consuming processing.

Previous studies have successfully applied deep learning models, including CNN- and YOLO-based approaches, to agricultural image

analysis, particularly for plant disease detection[1,2]. More recently, research has expanded toward fruit detection and ripeness assessment, demonstrating the potential of YOLO-based models for maturity classification and UAV-assisted monitoring under orchard conditions. Nevertheless, studies integrating close-range UAV image acquisition with real-time Hass avocado ripeness assessment remain limited. The present work addresses this gap by integrating UAV-based image acquisition with YOLO-based detection and classification models for real-time Hass avocado ripeness assessment. The objectives of this work are:

1.1. Traditional detection approaches

Computer vision is a branch of artificial intelligence that allows computers to automatically interpret and understand images and videos. Through mathematical algorithms and learning models, it is possible to

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extract relevant information from the visual environment, such as colors, shapes, textures, objects, and patterns. Its applications in agriculture include fruit classification, maturity detection, yield counting, disease monitoring, and process automation. In crops such as Hass avocados, it allows for the evaluation of visible characteristics (color, texture, size) without physical contact, reducing subjectivity and costs [3,4].

1.2. Computer vision and deep learning approaches for detection

An irrigation control system based on multilayer artificial neural networks with a backpropagation algorithm was developed for a micro-greenhouse dedicated to basil cultivation [5]. The system integrates electronic instrumentation and image analysis to monitor environmental and physiological variables, including ambient temperature and humidity, as well as individual plant characteristics. By processing data obtained from sensors and computer vision, the model enables automated irrigation based on plant status, optimizing water use and improving crop control. This work demonstrates the potential of neural networks in smart agricultural applications focused on crop automation and monitoring.

A systematic literature review was conducted on the use of convolutional neural networks (CNNs) and deep learning techniques for detecting diseases in tuber leaves [6]. Following Kitchenham's methodology for systematic reviews, the authors applied inclusion and exclusion criteria, as well as quality assessments, selecting 35 relevant studies for analysis. The results show that CNN-based models and transfer learning achieve high levels of accuracy in plant disease classification tasks, highlighting the importance of image preprocessing and dataset quality for model performance. Convolutional Neural Networks (CNNs) have become one of the most widely used architectures for image classification due to their ability to automatically learn hierarchical visual features from large-scale datasets [7–9]. Recent studies have further demonstrated the applicability of optimized CNN architectures for automated plant image classification and the importance of data augmentation for improving classification performance [10].

A deep learning approach for detecting diseases in potato leaves using convolutional neural networks (CNNs) was proposed [11]. The model was trained on a dataset of labeled images representing different crop health conditions, allowing the network to automatically learn features relevant to classification. The results demonstrated that CNNs are an effective tool for diagnosing plant diseases from images, improving accuracy and efficiency compared to traditional methods. This study reinforces the applicability of deep learning in agricultural systems based on machine vision.

In recent studies, advanced deep learning-based approaches have been proposed for identifying diseases in potato leaves. For example, a model called SAM-CNNNet was developed, which incorporates a spatial attention mechanism and hyperparameter optimization using Elephant Grazing Optimization (EHO) [12]. The model was evaluated using the PlantVillage dataset, achieving an accuracy of 98.58%, demonstrating the effectiveness of the proposed approach for precision agriculture applications.

Recent studies have demonstrated the effectiveness of deep learning techniques for fruit detection and yield estimation in agricultural settings [13]. Furthermore, UAV-based monitoring systems have been successfully applied to fruit detection in orchards [14]. Object detection has evolved significantly with the development of region-based and single-stage detectors, including Faster R-CNN, SSD, and YOLO architectures [15–18]. Deep learning models, particularly convolutional neural networks, have also been applied to classify fruit maturity stages [19]. More recently, YOLO-based approaches have emerged as effective solutions for real-time fruit detection and ripeness assessment [20].

Recent advances have further extended these approaches to fruit maturity analysis. Xiao et al. [21] employed a YOLOv8-based model for fruit ripeness identification, demonstrating that modern object detection

architectures can effectively discriminate among different maturity stages. Similarly, Liang et al. [22] combined UAV remote sensing with an improved YOLOv8 model to detect litchi fruit maturity under orchard conditions, highlighting the potential of UAV-based image acquisition for automated maturity assessment. In addition, Kamat et al. [23] compared YOLO and SSD models for multi-class fruit ripeness detection, confirming the effectiveness of deep learning approaches for distinguishing multiple maturity stages. Collectively, these studies demonstrate the growing interest in automated fruit maturity assessment. However, applications integrating close-range UAV imagery with real-time Hass avocado ripeness classification remain limited, motivating the development of the proposed system.

1.3. Knowledge gap and study justification

Despite significant advances in deep learning for agricultural applications, most existing studies focus on disease detection or monitoring in controlled environments. Furthermore, while UAV-based systems and YOLO architectures have shown promising results for fruit detection, research on the combined problem of real-time detection and maturity classification under field conditions is limited. Object detection has evolved considerably over the last decade, progressing from region-based detectors to real-time single-stage architectures. Faster and more efficient models have enabled practical deployment in agricultural monitoring applications, particularly in scenarios involving dense fruit distributions, occlusion, and computational constraints [24,25]. Among these approaches, the YOLO family has become one of the most widely adopted solutions due to its balance between detection accuracy and computational efficiency [24,26–28].

In this context, this study proposes an integrated approach that combines UAV data acquisition with deep learning models for the detection and classification of avocado maturity stages, aiming to provide a practical solution for agricultural monitoring under real-world conditions.

To address this gap, the objectives of this study were:

1. To develop a UAV-based real-time monitoring system for Hass avocado orchards.
2. To implement a YOLO-based model for avocado fruit detection under real-world field conditions.
3. To classify avocado maturity into five ripeness stages using a deep learning classification model.
4. To experimentally evaluate the proposed system under challenging field conditions, including varying illumination and partial fruit occlusion.

2. Materials and methods

Cultures, video recording and analysis.

3. Dataset preparation

Initially, the Hass Avocado Ripening Photographic Dataset was used, which contains 14,710 images of Hass avocados classified into five ripeness levels: unripe, midripe, ripe, fullripe, and overripe. The initial ripeness dataset was based on the Hass Avocado Ripening Photographic Dataset [29]. This dataset was automatically organized using a script to structure the images into folders corresponding to each category.

In the first stage, this dataset was used to train a model based on the YOLO architecture, aiming to simultaneously detect fruits and classify their ripeness stage. However, during preliminary tests, limitations were identified in terms of processing speed and frames per second (FPS), which hindered its application in real-time scenarios.

For this reason, a two-stage strategy was adopted: fruit detection and ripeness classification. For the detection stage, a new dataset was generated through Real World Adaptation, using images extracted from

videos captured in Hass avocado crops under real field conditions. From these videos, 340 images were selected and annotated with bounding boxes using the Roboflow platform. The dataset was divided into 280 images for training, 40 for validation, and 20 for testing. Additionally, data augmentation techniques were applied, including horizontal flipping, brightness variation ($\pm 15\%$), saturation adjustment ($\pm 15\%$), and exposure modification ($\pm 10\%$).

For the ripeness classification stage, the initial dataset of labeled fruits was reused. To increase the number of available samples, the regions corresponding to each detected fruit in the annotated images were automatically cropped. Since a single image can contain multiple fruits, this procedure allowed multiple samples to be generated from the same image.

As a result, an expanded dataset of 20,231 avocado images was obtained, of which 17,237 were used for training and 2994 for validation, distributed across the five ripeness categories. This approach improved the dataset's representativeness and enhanced the model's generalization capability under real field conditions.

Table 1 provides a summary of the datasets used throughout this study. It includes the original ripening dataset, the detection dataset derived from real field conditions, and the final expanded classification dataset obtained after the cropping and augmentation processes.

As shown in Table 1, the proposed methodology significantly increases the number of samples available for classification, enhancing the diversity and robustness of the dataset.

The ripeness labels used in this study were adopted from the original Hass Avocado Ripening Photographic Dataset. The dataset was constructed using 478 Hass avocados acquired three days after harvest and stored under three temperature conditions (10°C , 20°C , and ambient temperature) throughout the ripening process. Each fruit was photographed daily under controlled conditions, and every image was assigned to one of five ripening stages defined by the dataset authors: Underripe, Breaking, Ripe (First Stage), Ripe (Second Stage), and Overripe. For consistency and readability throughout this study, these labels were renamed as Unripe, Midripe, Ripe, Fullripe, and Overripe, respectively. These predefined labels were used as the ground truth for training and evaluating the ripeness classification model.

Table 2 shows the distribution of images used for training and validation across the five ripeness categories, providing an overview of the dataset balance for the classification task.

As shown in Table 2, the dataset exhibits a relatively balanced distribution among the five ripeness categories, with a higher number of samples in the fullripe and unripe classes. This balance is important to ensure that the classification model does not become biased toward a specific category.

Although slight variations exist between classes, the number of training and validation samples remains sufficiently representative for each category, enabling the model to learn discriminative features across different ripeness stages. This distribution contributes to the robustness and generalization capability of the classification model.

3.1. Evaluation metrics in object detection

The evaluation of object detection models is based on standardized metrics that quantify both localization accuracy and the model's classification capability. Among the most commonly used metrics are precision, recall, Intersection over Union (IoU), Average Precision (AP), mean Average Precision (mAP), and the F1-score. To evaluate object

Table 1
Dataset summary.

Dataset	Images	Description
Avocado Ripening Dataset	14,710	Manually labeled dataset
Detection Dataset	340	Images extracted from field videos
Final Classification Dataset	20,231	Cropped fruit samples

Table 2
Images used for training.

Class	Training	Validation
Unripe	3840	717
Midripe	2482	390
Ripe	3119	538
Fullripe	4084	699
Overripe	3712	650

detection performance, standard benchmarks and metrics established by the PASCAL VOC and Microsoft COCO datasets are commonly employed. These benchmarks introduced evaluation criteria based on Intersection over Union (IoU), Average Precision (AP), and mean Average Precision (mAP), which remain the standard metrics for assessing detection models [30,31].

Precision measures the proportion of correct detections (true positives) relative to the total number of detections made by the model, indicating the level of false positives. Recall, on the other hand, measures the proportion of objects correctly detected relative to the total number of actual objects present in the image, reflecting the model's ability to avoid missed detections.

Intersection over Union (IoU) quantifies the degree of overlap between the predicted bounding box and the ground truth box. It is calculated as the ratio between the area of intersection and the area of union of both boxes. A minimum IoU threshold (e.g., 0.5) is typically used to determine whether a detection is considered valid.

Average Precision (AP) is obtained from the area under the Precision-Recall curve for a specific class, while mean Average Precision (mAP) corresponds to the average of AP values across all classes. In modern evaluation protocols, the metric mAP@0.5:0.95 is commonly used, as it averages performance over multiple IoU thresholds (from 0.5 to 0.95 in increments of 0.05), providing a more comprehensive and stringent assessment of model performance.

Finally, the F1-score is defined as the harmonic mean of precision and recall, making it particularly useful in scenarios with class imbalance, as it penalizes both false positives and false negatives.

3.2. Relevance in agricultural applications

In precision agriculture contexts, evaluation metrics must be interpreted considering real field conditions, such as variations in lighting, partial occlusions by leaves, shadows, and variability in fruit size and orientation where conditions are not controlled [1]. These factors introduce significant challenges for computer vision models, particularly in outdoor environments where conditions are not controlled. In this scenario, the mAP@0.5:0.95 metric is particularly relevant, as it simultaneously evaluates precise localization capability and the model's robustness under different spatial constraints and IoU thresholds. This provides a more comprehensive assessment of model performance compared to single-threshold metrics.

Similarly, the F1 score becomes especially important when there is class imbalance, for example, when the number of ripe fruits is significantly lower than that of green fruits, a common situation in phenological monitoring and harvesting stages. In such cases, relying solely on accuracy may lead to misleading conclusions, whereas the F1 score offers a balanced measure by considering both precision and recall. This allows for a more reliable evaluation of the model's effectiveness in detecting and correctly classifying fruits under real agricultural conditions.

3.3. Data acquisition tool

Scrcpy is an open-source tool that enables users to display and control an Android device directly from a computer without requiring the installation of additional applications on the mobile device. It operates

as an efficient screen mirroring system, providing low-latency transmission and high visual quality.

The tool functions by capturing the content displayed on the smartphone screen, encoding the video stream using efficient compression codecs such as H.264 or H.265, and transmitting it via a USB or wireless connection to the computer. The received stream is then decoded and rendered in real time within a dedicated window on the host device.

One of the main advantages of Scrcpy is its ability to maintain high frame rates with minimal delay, making it suitable for real-time applications such as remote monitoring and computer vision processing. In this study, Scrcpy plays a key role in enabling the transmission of live video captured by the UAV-mounted camera—via the mobile device—to the laptop, where detection and classification algorithms are executed. This integration allows for real-time analysis of agricultural data under field conditions without requiring specialized transmission hardware.

3.4. Model development and training

Deep learning techniques were implemented using two independently trained but complementary models based on the Ultralytics YOLOv8 framework: an object detection model for fruit localization and an image classification model for ripeness estimation. Although both models belong to the YOLOv8 family and are based on convolutional neural networks (CNNs), they perform different computer vision tasks and were trained using different datasets.

For fruit detection under real orchard conditions, the Ultralytics YOLOv8n object detection architecture (Ultralytics version 8.3.227) was employed. A dedicated real-world dataset was created by extracting frames from UAV videos acquired in commercial Hass avocado orchards. The extracted images were manually annotated using Roboflow, where each avocado fruit was labeled with a bounding box and exported in YOLO format. The detector was initialized from the pretrained yolov8n.pt weights and trained for 50 epochs using an input image size of 640×640 pixels and a batch size of 16. Automatic optimizer selection was used with an initial learning rate of 0.01, momentum of 0.937, and weight decay of 0.0005. The detector performance was evaluated using

Precision, Recall, F1-score, mAP@0.5, mAP@0.5:0.95, and Precision–Recall curves.

For ripeness classification, the Ultralytics YOLOv8n-cls image classification architecture (Ultralytics version 8.3.227) was employed. The classifier was initialized from the pretrained yolov8n-cls.pt weights and trained using a dataset containing 17,237 training images and 2994 validation images, corresponding to five ripeness categories: Unripe, Midripe, Ripe, Fullripe, and Overripe. Training was performed for 200 epochs using RGB images resized to 224×224 pixels, a batch size of 64, automatic optimizer selection, an initial learning rate of 0.01, momentum of 0.937, and weight decay of 0.0005. The classification performance was assessed using Top-1 accuracy, training and validation loss, and the confusion matrix.

The proposed methodology integrates both models into a sequential workflow. First, the YOLOv8n detector localizes avocado fruits in each UAV image by predicting bounding boxes and confidence scores. Subsequently, each detected fruit is cropped and processed by the YOLOv8n-cls classifier, which assigns one of the five predefined ripeness categories. This two-stage architecture combines accurate fruit localization with ripeness estimation, enabling real-time monitoring of Hass avocado maturity in orchard environments.

Fig. 1 illustrates the complete workflow of the proposed system. During UAV operation, aerial images and videos are captured and transmitted in real time to a mobile device through the UAV communication link. The mobile device screen is mirrored to a laptop using scrcpy, allowing real-time image processing. The YOLOv8n detector first identifies individual avocado fruits in each frame, after which the YOLOv8n-cls classifier estimates the ripeness stage of every detected fruit. Finally, each fruit is labeled as Unripe, Midripe, Ripe, Fullripe, or Overripe, and the results are displayed together with the corresponding bounding boxes on the laptop screen for real-time monitoring and analysis.

3.5. Study area location

As shown in Fig. 2, the study area is located in the district of Coalque, General Sánchez Cerro province, Moquegua region, Peru,

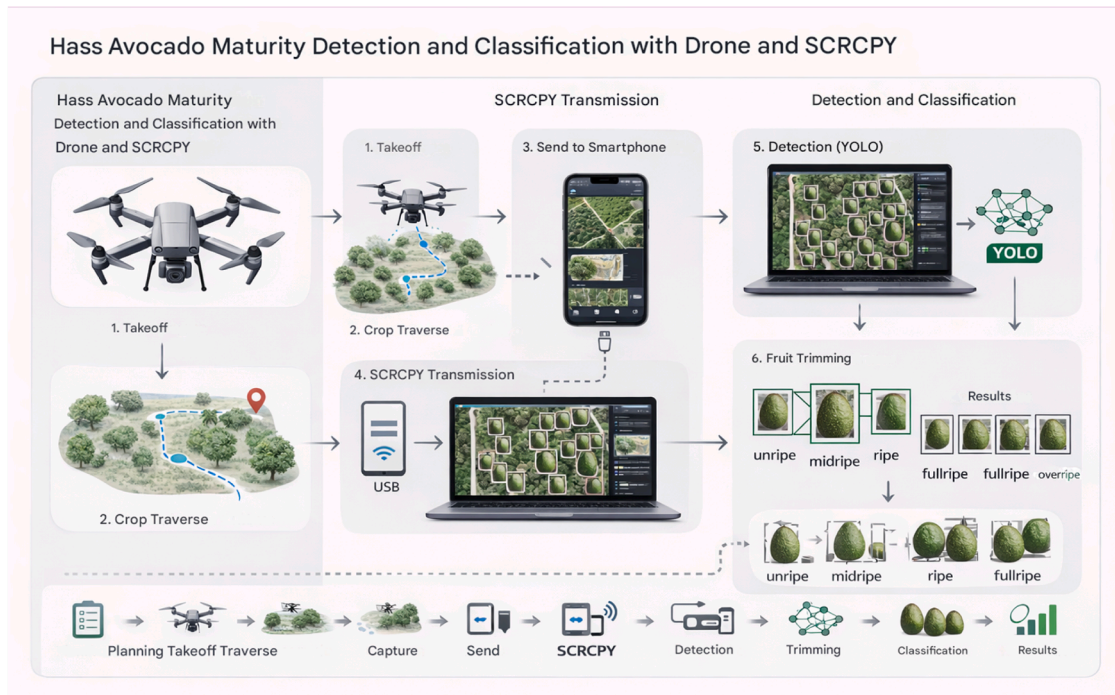


Fig. 1. Workflow of the proposed system for UAV-based detection and ripeness classification of Hass avocados using SCRCPY transmission and deep learning models.



Fig. 2. Location of the study area in Coalaque, Moquegua, Peru.

approximately at coordinates $16^{\circ}38'35.6''\text{S}$ and $71^{\circ}02'23.7''\text{W}$. This area is characterized by the agricultural production of Hass avocados.

3.6. Data collection using the Xiaomi FIMI X8 drone

The images used in this study were captured using a Xiaomi FIMI X8 SE unmanned aerial vehicle (UAV), equipped with a high-resolution camera capable of recording video in 4 K quality (3840×2160 pixels). The drone features a three-axis gimbal stabilization system, which allows for images with less vibration and greater clarity.

During data acquisition, the UAV flew over the Hass avocado crop along flight paths designed to cover the area of interest. The images were derived from videos captured under real-world field conditions, including variations in lighting, shadows, partial occlusion of fruit, and the presence of a natural background.

The video was streamed to a mobile device, which was then connected to a computer using the *scrcpy* tool, allowing the images to be viewed and processed in real time.

As shown in Fig. 3, the real-time transmission and execution of the Python program with both models loaded; it can be seen that the program detects the fruits and classifies them according to their ripeness, and the FPS is optimized for smooth transmission.

This approach enables the simulation of a real-time monitoring environment, integrating data capture via UAV with processing based on computer vision techniques.

Transmit real-time drone footage for analysis. Additionally, for real-time transmission with the Xiaomi FIMI X8 SE drone, the Waypoints feature (also known as Flight Plan or Waypoint Flight) is a flight mode in which you mark several points on the map within the app, and the drone automatically flies along that route. Before or during the flight, in the FIMI Navi app, you select the locations the drone will pass over on the map and define the altitude and parameters: You set the altitude for each point, speed, camera direction, and even whether it should take photos or record video at certain points—in this case, video recording. Once the mission begins, the drone flies autonomously from one waypoint to the next in the specified order, adjusting altitude and speed as programmed. Upon completing the route, you can instruct it to return to the starting point (Return to Home) or to remain hovering at the last point. This allows you to focus on controlling the camera, as the drone flies on its own, making it a highly effective model for this work in real-time detection of diseases in leaves and fruits.

As shown in Fig. 4, the image acquisition process was conducted

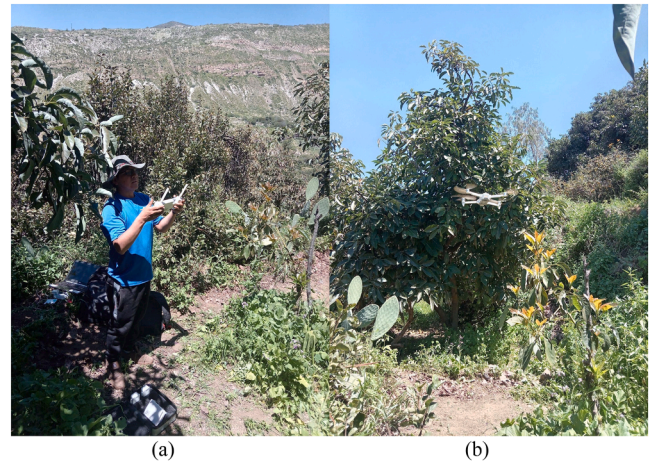


Fig. 4. Image acquisition process in the avocado field: (a) operator preparing the UAV; (b) UAV capturing images over the crop.

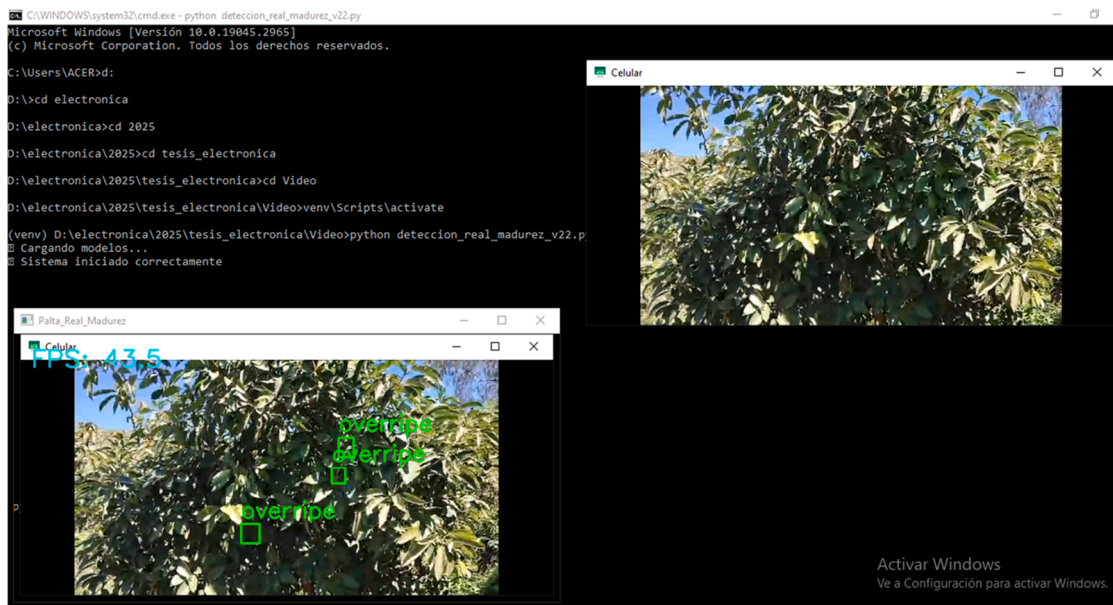


Fig. 3. Real-time transmission via *SCRCPY* and execution of the Python program.

under real field conditions in avocado crops. The workflow includes the preparation and operation of the UAV, followed by aerial image capture over the cultivation area. This process ensures that the collected data accurately reflect real environmental conditions, which is essential for improving the robustness of the proposed model.

Table 3 presents the main specifications of the UAV used for data acquisition, including key technical parameters relevant to image capture under field conditions.

As shown in Table 3, the UAV is equipped with a high-resolution 4 K camera and a 3-axis gimbal stabilization system, enabling stable and high-quality image acquisition. These features are essential for capturing reliable data under real field conditions, supporting the performance of the proposed detection and classification models.

Image acquisition was performed during close-range UAV inspection. Based on the geometry of the recorded 4 K video (3840×2160 pixels), the known dimensions of Hass avocados (approximately 12 cm in length), the apparent fruit size in the images (approximately 180–250 pixels per fruit), and the optical characteristics of the FIMI X8 SE camera, the working distance between the camera and the fruit canopy was estimated to be approximately 1.5–2.5 m. Considering the typical height of commercial Hass avocado trees under pruning management (approximately 3–4 m), the UAV operated at an estimated flight altitude of about 2–3 m above ground level. Under these close-range acquisition conditions, each fruit occupied sufficient image resolution to preserve color and texture information required for ripeness classification.

3.7. Yolo model training

The lightweight variant of the YOLOv8 architecture (YOLOv8n) was used due to its computational efficiency and suitability for real-time applications.

The model was trained using the real-world dataset described above, which contains images annotated with bounding boxes corresponding to Hass avocados. The training was conducted using an input image size of 640×640 pixels.

The training process was carried out using the implementation provided by the Ultralytics YOLO library, with the training configuration described in Section 2.5. During training, performance metrics such as precision, recall, and mean average precision (mAP) were monitored to evaluate the model's ability to correctly detect the objects of interest.

To analyze the model's performance, integrated visualization tools such as TensorBoard and the Ultralytics results interface were used, enabling the tracking of the convergence process and the identification of potential overfitting issues.

3.8. Classification model

A classification model was implemented to determine the ripeness level of each detected avocado. The model takes as input the cropped regions obtained during detection and generates as output one of five predefined classes: unripe, midripe, ripe, fullripe, and overripe.

The model was trained using the generated classification dataset, consisting of 20,231 images. Each image corresponds to a cropped avocado image obtained from the detection dataset, allowing the model to focus exclusively on the fruit region.

The classification process runs after detection, forming a two-stage

workflow in which object localization and ripeness estimation are performed sequentially.

3.9. Real-time implementation

The real-time system was implemented by connecting a mobile device to a laptop using SCRCPY, a method for transmitting data from a cell phone to a laptop. The cell phone, which is connected to the drone controller via a USB cable and the FIMI NAVY app, receives the video captured by the camera of the Xiaomi FIMI X8 SE drone. This real-time transmission is displayed in a window where the code running the detection and classification models is executed, and as a result, the Hass avocados are shown enclosed in bounding boxes, classified, and displayed at high FPS.

3.10. Python program

The system was implemented in Python using computer vision libraries such as OpenCV and real-time screen capture tools via mss. Detection and classification were performed using models based on Ultralytics YOLO.

Processing is performed on frames captured from the mobile device's screen using scrcpy. Each frame is initially analyzed by the detection model, which identifies the fruits using bounding boxes. Subsequently, the detected regions are automatically cropped and processed by a classification model that determines the ripeness level.

To improve real-time performance, optimization strategies were implemented, such as processing alternate frames and maintaining detection boxes over time. The latter allows detected boxes to be retained across several consecutive frames, even in the absence of new detections, reducing visual instability and improving the system's performance.

3.11. Optimization for real-time processing

To ensure the system operates in real time, optimization strategies were implemented to reduce computational load and improve the visual stability of detections.

First, a *frame-skipping* technique with a skip of 2 (PROCESS_EVERY = 2) was applied, in which only certain video frames are processed at defined intervals, rather than analyzing all captured frames. This strategy reduces the processing time per frame and increases the frames per second (FPS), while maintaining adequate system performance.

Additionally, an 8-frame detection persistence mechanism (PERSIST_FRAMES=8) was incorporated, which allows bounding boxes and classification labels to be maintained across several consecutive frames, even when no new objects are detected in a specific frame. This technique reduces the flickering effect in detections and improves the system's visual stability under real-time processing conditions.

Finally, the system follows a sequential pipeline in which each captured frame is processed using a YOLO-based detection model, followed by the cropping of the regions of interest corresponding to the detected fruits, and subsequently the classification of the ripeness status using an independent model.

The general system workflow can be summarized as:

**Frame → Detection
(YOLO) → Cropping → Classification → Visualization**

4. Results and experiments

4.1. Detection metrics

The detection model was developed using the YOLOv8 architecture, designed to identify avocados under real-world field conditions. The detection task was formulated as a single-class problem, corresponding

Table 3
Specifications of the UAV used.

Parameter	Specification
Model	Xiaomi FIMI X8 SE
Camera resolution	4 K (3840×2160)
Stabilization	3-axis gimbal
Capture type	Video
Environment	Real-world field conditions

to the “avocado” category.

The main purpose of the model was to locate the fruits in images captured via real-time transmission from the drone, using SCRCPY as the visualization interface. The regions detected using bounding boxes were subsequently used as input for the ripeness classification model.

The detection model presented in Table 4 achieved a precision of 0.7951 and a recall of 0.7849, indicating a balanced performance between false positives and false negatives. The mAP@0.5 reached 0.8388, demonstrating strong object detection capability under standard evaluation conditions. Furthermore, the mAP@0.5:0.95 value of 0.6308 reflects consistent performance under stricter IoU thresholds.

Although the mAP@0.5 is slightly below 0.85, the model maintains robust performance in real-world scenarios, particularly under challenging conditions such as variations in illumination and partial occlusion. These results indicate that the model generalizes effectively and is capable of detecting avocado fruits even in complex field environments.

Overall, the results summarized in Table 4 confirm the effectiveness of the proposed model, demonstrating its capability to perform accurate and robust avocado detection under real field conditions.

In Fig. 5, the area under the curve (AUC) for the *avocado* class is 0.843, and the overall performance achieves an mAP@0.5 of 0.843. The curve illustrates the trade-off between precision and recall at different confidence thresholds, with high precision in mid-range recall values and a progressive decrease as detection coverage increases.

In Fig. 6, the maximum F1 score is reached around a threshold of 0.23 ($F1 \approx 0.80$), indicating the best trade-off between precision and recall. For high thresholds, F1 decreases due to the loss of detections (reduced recall), while for very low thresholds, the false positive rate increases (reduced precision).

In Fig. 7, accuracy increases as the threshold is raised, reaching values close to 1.0 for high thresholds (≈ 0.88), indicating a significant reduction in false positives. However, high thresholds result in a decrease in recall; therefore, the selection of the threshold must balance accuracy and detection coverage depending on the application scenario.

In Fig. 8 shows a progressive decrease in recall as the confidence threshold increases, with a sharp drop for high values ($\approx 0.8-1.0$), reflecting the loss of true detections when applying stricter criteria. This highlights the trade-off between detection coverage and false positive reduction when selecting the operating threshold.

Fig. 9 shows the correct predictions and classification errors between the target class and the *background*. The main diagonal represents the correct predictions, while the values outside the diagonal correspond to errors.

Fig. 10. The training and validation results of the detection model show a stable and consistent convergence across epochs. The loss functions, including box loss, classification loss, and distribution focal loss (DFL), exhibit a progressive decrease during training, indicating effective feature learning and optimization. In parallel, the evaluation metrics, such as precision, recall, and mean Average Precision (mAP@0.5 and mAP@0.5:0.95), show a steady increase, reaching values close to 0.8–0.9, which reflects a strong detection capability of the model.

Furthermore, the similarity between training and validation curves suggests that the model generalizes well to unseen data, with no evident signs of overfitting or divergence. Minor fluctuations observed in validation metrics can be attributed to variations in the dataset and

Table 4
Detection metrics.

Metric	Value
Epoch	50
Precision (P)	0.7951
Recall (R)	0.7849
mAP@0.5	0.8388
mAP@0.5:0.95	0.6308

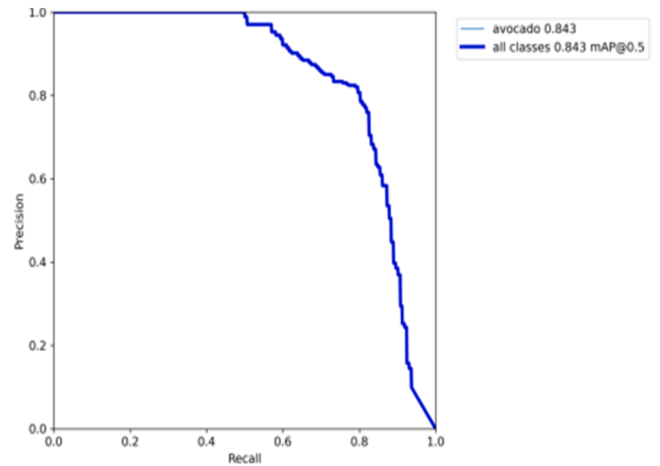


Fig. 5. Precision–Recall (PR) curve for the avocado detection model.

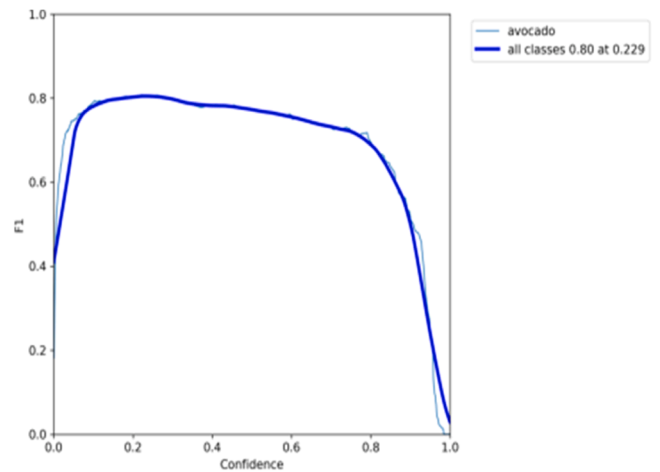


Fig. 6. F1 curve as a function of the confidence threshold of the avocado detection model.

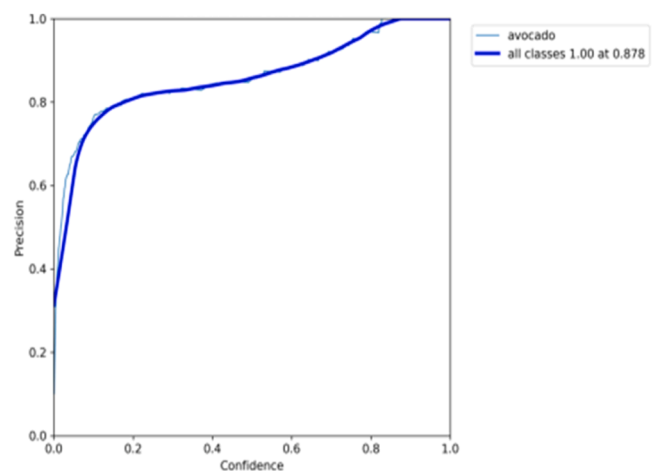


Fig. 7. Precision curve as a function of the confidence threshold of the avocado detection model.

environmental conditions during image acquisition. Overall, these results confirm that the model achieves reliable performance and stable learning behavior throughout the training process.

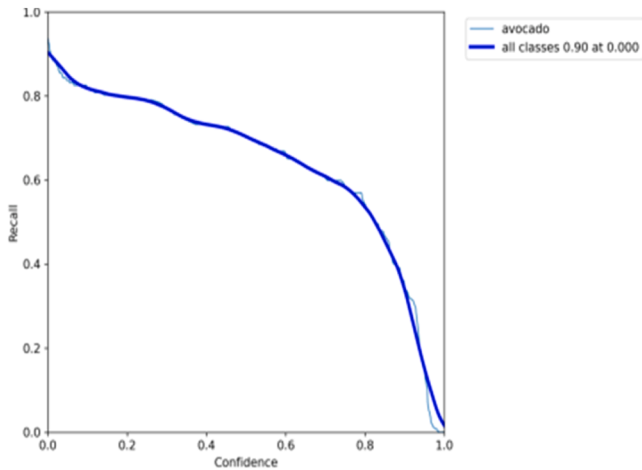


Fig. 8. Recall curve as a function of the confidence threshold of the avocado detection model.

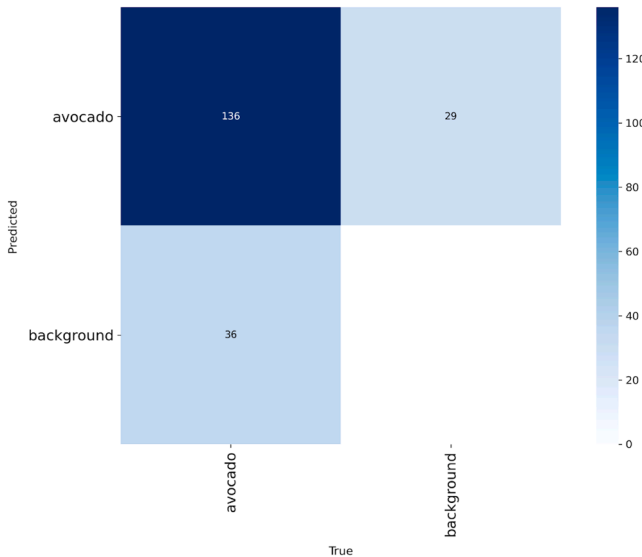


Fig. 9. Confusion matrix for the detection model for the "avocado" class.

4.2. Classification metrics

As shown in Table 5, the model achieves its best performance on the unripe class, with an F1-score of 0.91, while lower performance is observed for the midripe class, which reaches an F1-score of 0.67. This difference can be attributed to the higher visual similarity between intermediate ripeness stages, which makes classification more challenging.

Overall, the model demonstrates balanced performance across all classes, achieving a weighted F1-score of 0.78. These results indicate that the proposed approach is capable of effectively distinguishing between different ripeness stages under real-world conditions.

Additionally, the relatively high recall observed in the unripe class suggests that the model is less likely to miss early-stage fruits, which is particularly important for monitoring and decision-making in agricultural applications.

The classification model presented in Table 6 achieved a Top-1 accuracy of 0.78 and an overall accuracy of 0.79, indicating solid performance across the five ripeness categories. The Top-5 accuracy reached 1.00, suggesting that the correct class is consistently included among the top predictions.

As shown in Table 6, the model achieves a macro-average F1 score of

0.77 and a weighted average F1 score of 0.78, indicating balanced performance across classes despite variations in sample distribution. The high Top-5 accuracy further confirms the model's robustness, demonstrating its ability to reliably identify the correct ripeness stage even when the top prediction is uncertain.

Overall, these results validate the effectiveness of the proposed classification approach for practical applications in avocado ripeness assessment under real-world conditions.

In Fig. 11, the values on the main diagonal represent correct predictions per class, while the values off the diagonal indicate confusion between adjacent ripeness states. Greater overlap is observed between contiguous classes (e.g., ripe–fullripe and fullripe–overripe), consistent with the visual similarity between ripeness stages.

Fig. 12 shows the training curves of the classification model over 200 epochs. The training loss decreases steadily throughout the optimization process, indicating effective learning of the model parameters. In contrast, the validation loss decreases during the initial epochs but gradually increases after approximately the first 40–50 epochs, suggesting the onset of moderate overfitting while the training accuracy continues to improve.

The observed overfitting can be explained by the high visual similarity among adjacent ripeness stages and by the limited variability of the available dataset. As training progresses, the model becomes increasingly specialized to the training samples, resulting in lower training loss without a corresponding improvement in validation performance. Although data augmentation and regularization techniques were applied during training, the validation loss indicates that additional regularization or earlier stopping could further improve the model's generalization capability.

Regarding class-wise performance, the midripe class exhibited the lowest classification accuracy, as also observed in the confusion matrix (Fig. 11). Most misclassifications occurred between neighboring ripeness stages (unripe and ripe), reflecting the gradual physiological transition of avocado maturation rather than abrupt visual changes. During this intermediate stage, differences in skin color, texture, and illumination are subtle, making the class boundaries less distinguishable even for human observers. In contrast, the extreme classes (unripe and overripe) present more distinctive visual characteristics, resulting in higher classification accuracy.

Despite these challenges, the top-1 accuracy stabilizes at approximately 78%, while the top-5 accuracy remains close to 100% throughout training, indicating that the correct ripeness stage is almost always included among the model's highest-confidence predictions. Overall, the results demonstrate that the proposed classifier learned meaningful visual representations of avocado ripeness while highlighting the inherent difficulty of discriminating intermediate maturity stages.

Fig. 13 shows qualitative detection results obtained under real field conditions. The model successfully identifies avocados at different maturity stages, including unripe, ripe, and overripe, despite variations in illumination, occlusions caused by leaves, and complex backgrounds. The predicted bounding boxes and class labels demonstrate the model's ability to localize and classify multiple fruits within the same scene.

Additionally, the results highlight the robustness of the proposed approach when applied to natural agricultural environments, where factors such as shadows, perspective changes, and varying fruit sizes can affect detection performance. These qualitative outcomes complement the quantitative metrics presented earlier, confirming the practical applicability of the model for real-world avocado monitoring tasks.

5. Discussion

Fig. 9 presents the confusion matrix of the model for detecting the avocado class. There are 136 true positives (correctly detected fruits), 29 false positives (incorrect detections classified as avocado), and 36 false negatives (undetected fruits). The highest concentration of values is found on the main diagonal, indicating adequate model performance in

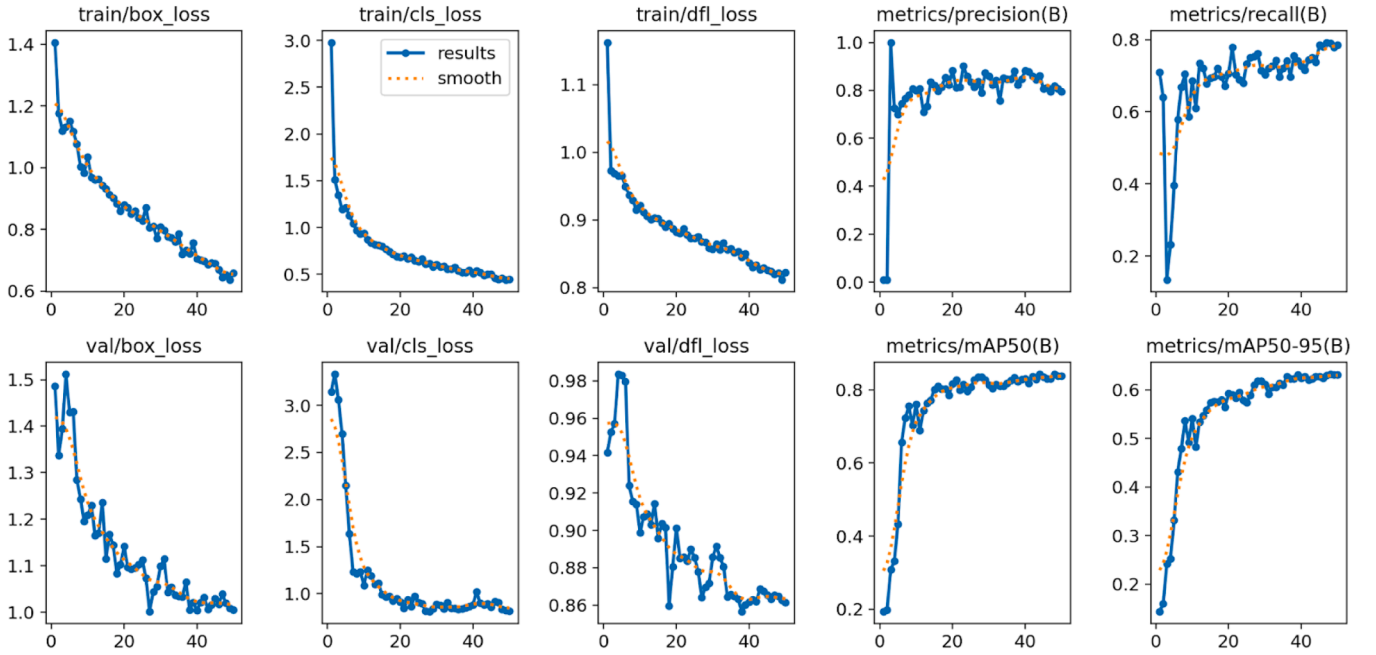


Fig. 10. Training and validation metrics of the detection model over epochs, including box loss, classification loss, distribution focal loss (DFL), precision, recall, and mAP@0.5 and mAP@0.5:0.95.

Table 5

Classification model metrics.

Class	Precision	Recall	F1-score	Samples
Unripe	0.87	0.95	0.91	717
Midripe	0.73	0.63	0.67	390
Ripe	0.78	0.76	0.77	538
Fullripe	0.7	0.76	0.73	699
Overripe	0.82	0.75	0.78	650

Table 6

Global metrics for classification.

Metric	Value
Accuracy	0.79
Macro Avg F1	0.77
Weighted Avg F1	0.78
Top-1 Accuracy	0.78
Top-5 Accuracy	1

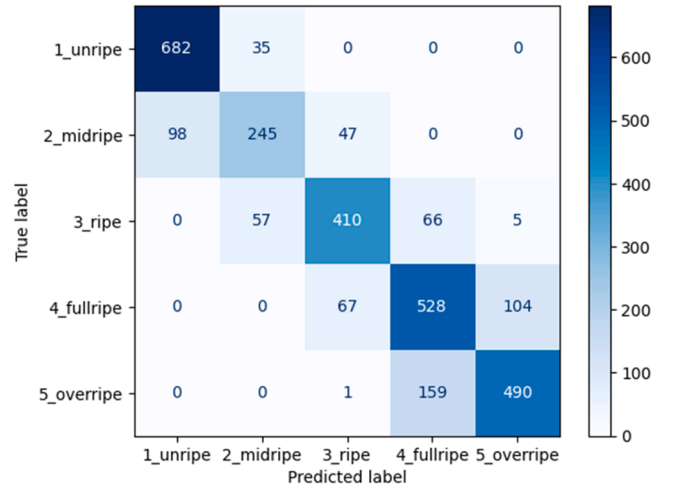


Fig. 11. Confusion matrix of the Hass avocado ripeness classifier for five classes (unripe, midripe, ripe, fullripe, and overripe).

and recall were calculated as follows:

$$Precision = \frac{136}{136 + 29} = 0.82 \quad (2)$$

$$Recall = \frac{136}{136 + 36} = 0.79 \quad (3)$$

The F1 score is then computed as:

$$F1 = \frac{2(0.82)(0.79)}{0.82 + 0.79} \approx 0.80 \quad (4)$$

distinguishing between object and background. However, the presence of false negatives suggests that there are still cases where the model misses fruits, possibly due to variations in lighting, occlusions, or chromatic similarities with the surroundings. In agricultural applications, this behavior can be prioritized by adjusting the confidence threshold to favor recall when it is critical to avoid missing ripe fruits.

The F1 vs. Confidence curve shows how the F1 score changes when you adjust the detector's confidence threshold

$$F1 = \frac{2 \cdot Precision \cdot Recall}{Precision + Recall} \quad (1)$$

The F1 score is defined as the harmonic mean of precision and recall, as shown in Eq. (1). According to the previous analysis, the maximum F1 score is approximately 0.80 with an optimal confidence threshold between 0.4 and 0.5.

This indicates that the model achieves a balance between precision (minimizing false positives) and recall (minimizing false negatives).

Based on the confusion matrix, the model yielded 136 true positives, 29 false positives, and 36 false negatives. From these values, precision

The values used in Eqs. (2)–(4) were obtained from the confusion matrix generated on the validation subset using the confidence threshold at which the maximum F1 score was achieved (approximately between 0.4 and 0.5).

This result confirms the balance between precision and recall, indicating stable detection performance under real field conditions. The F1

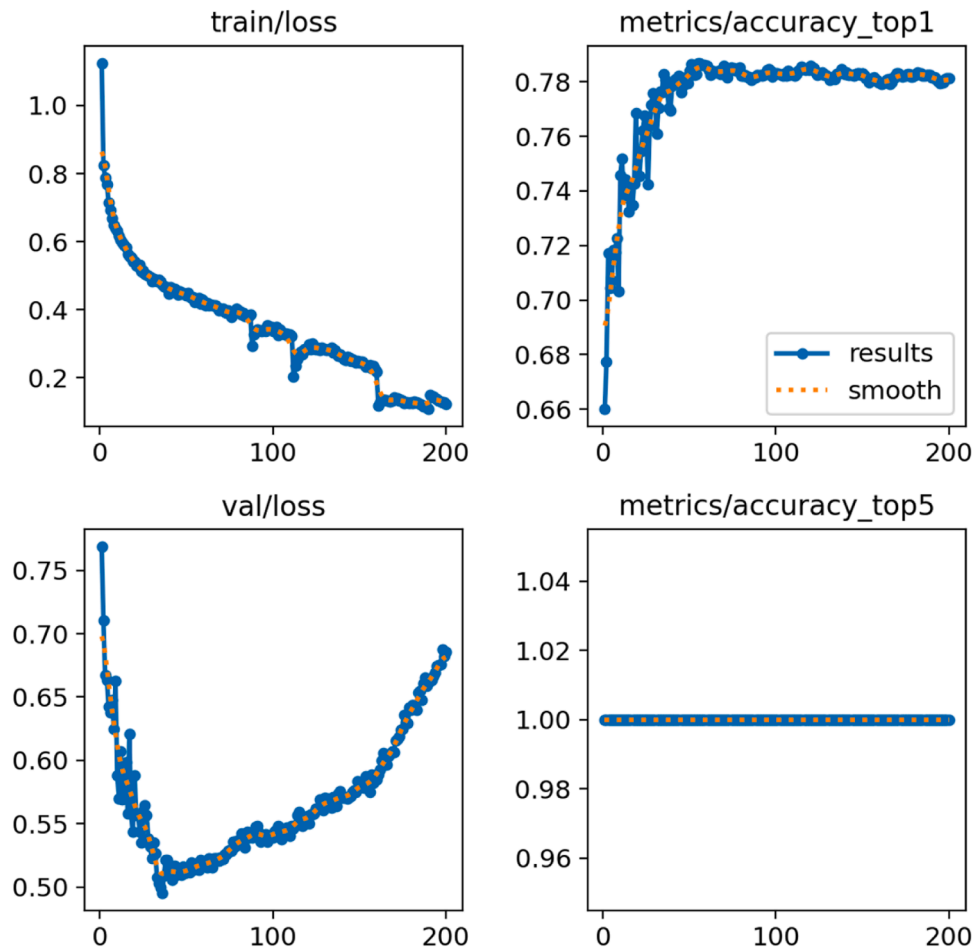


Fig. 12. Training curves for the classification model: training loss (top left), validation loss (bottom left), top-1 accuracy (top right), and top-5 accuracy (bottom right). The training error converges, and the top-1 accuracy stabilizes around 78%.

score obtained from the confusion matrix (≈ 0.80) matches the maximum value observed on the F1 curve versus confidence, confirming that the selected threshold corresponds to the model's optimal operating point. This agreement supports the consistency between the graphical analysis and the quantitative evaluation, reinforcing the reliability of the threshold selection process.

The precision (≈ 0.82) and recall (≈ 0.79) values indicate that the model is slightly biased toward precision, resulting in fewer false positives than false negatives. In agricultural applications, this behavior reduces the risk of overcounting fruits, although some may go undetected under challenging conditions such as partial occlusion or complex lighting. If maximizing harvest efficiency is required, a lower confidence threshold could be considered to increase recall at the expense of a slight reduction in precision. Similar CNN-based approaches have shown promising performance in agricultural image classification tasks [1,2,10].

As shown in Fig. 12, the training loss decreases progressively to values close to 0.12, indicating adequate convergence of the model. However, the validation loss begins to increase around epoch 60, suggesting the presence of overfitting, as the model continues improving on the training data while losing generalization capability. The top-1 accuracy stabilizes around 77.5%–78%, consistent with the confusion matrix results for the five ripeness classes, which is reasonable given the visual similarity between intermediate stages. The top-5 accuracy reaches 100%, as expected in a five-class classification problem, since the correct class is always included among the five most probable predictions.

The performance of the detection model was evaluated on an independent validation dataset using the standard Ultralytics YOLO

evaluation protocol. The reported Precision, Recall, mAP50, and mAP50–95 values were obtained by comparing the predicted bounding boxes with the corresponding manually annotated ground-truth labels. These metrics represent the final performance of the trained detector on the validation set and provide a quantitative assessment of its localization and detection capability under real-field conditions. As commonly reported in YOLO-based object detection studies, the evaluation results are presented as point estimates. Although statistical uncertainty measures such as confidence intervals or variance were not included in the present study, future work will consider repeated experimental evaluations and bootstrap-based statistical analysis to further quantify the variability of the reported detection performance.

6. Limitations

Although the proposed system achieved promising results under real field conditions, certain limitations remain. Detection performance may decrease under severe occlusion, extreme illumination conditions, or when fruits are partially hidden by foliage. In addition, the dataset size remains limited compared to large-scale agricultural datasets, which may affect generalization capability.

7. Conclusions

Detection is performed using a single-stage method with real-world detection followed by classification. Both models were trained on a large number of images, and the resulting data demonstrate high accuracy in detection results.

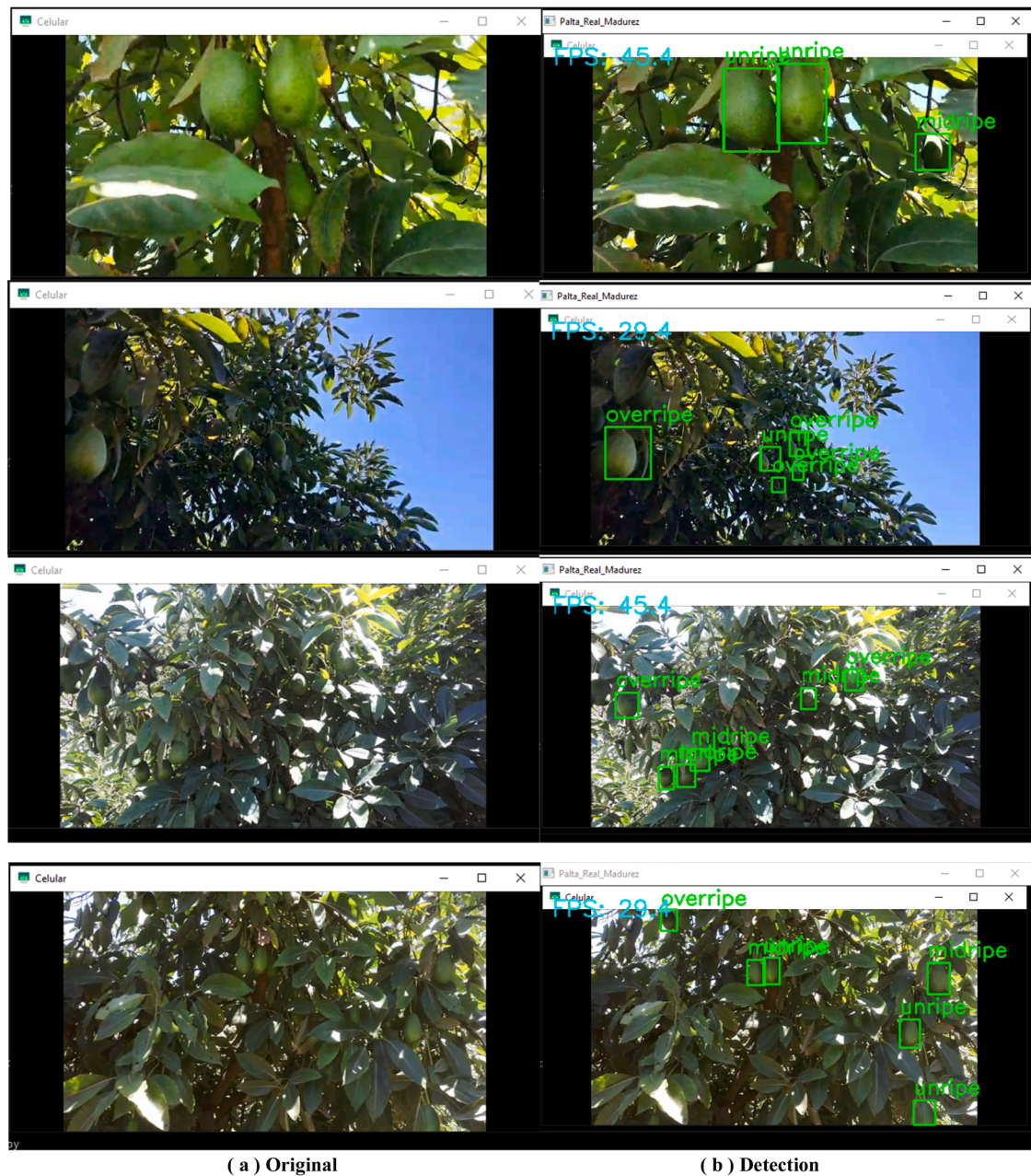


Fig. 13. Qualitative detection results under real field conditions. Each pair of images presents the original scene (left) and the corresponding detection output (right), where bounding boxes and predicted ripeness stages are displayed. The examples highlight the robustness of the model under challenging conditions, including illumination variability, partial occlusions, and complex natural backgrounds.

The proposed YOLO-based model demonstrated solid performance in fruit detection under real-world field conditions, achieving an accuracy of 79.51%, a recall of 78.49%, and a mAP@0.5 of 83.88%. These results demonstrate the system's adequate ability to identify and locate fruits with a considerable level of reliability. Furthermore, the mAP@0.5:0.95 of 63.08% indicates that the model maintains stable performance even under stricter evaluation criteria, although there is room for improvement in fine- localization accuracy.

In the ripeness stage classification phase, the model achieved an overall accuracy of 79%, with an outstanding F1-score of 0.91 for the "unripe" class. However, greater difficulty was observed in classifying intermediate stages such as "midripe," which can be attributed to visual similarities between consecutive phenological stages.

Overall, the results confirm the viability of the proposed approach for precision agriculture applications, particularly in UAV-based

monitoring systems. For future work, we recommend expanding the dataset, improving class balance, and exploring data augmentation techniques or model fine-tuning to enhance the discrimination between similar maturity states.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used ChatGPT (OpenAI) to assist in language improvement, grammar correction, and manuscript organization. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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CRediT authorship contribution statement

Verner Leonidas Tudela Taco: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Wilson Emerson Quispe Accostupa:** Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization. **Erasmus Sulla Espinoza:** Visualization, Formal analysis.

Declaration of competing interest

The authors declare no conflict of interest.

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Data availability

Data will be made available on request.

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